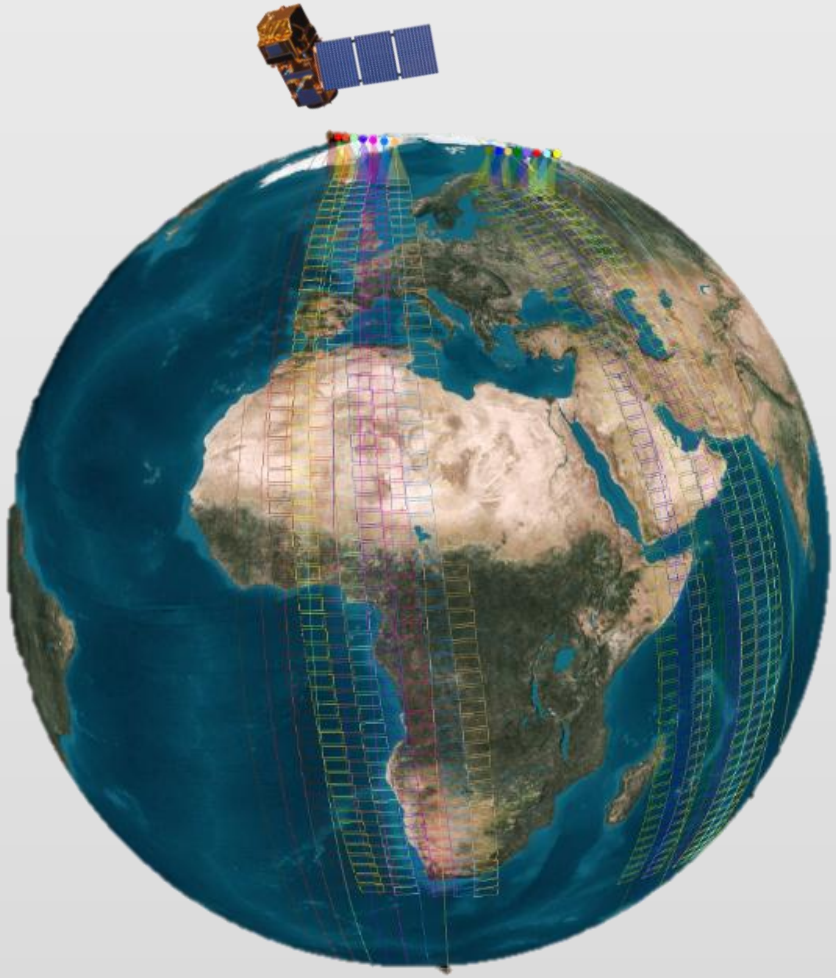
Pixel-based landslide detection from Sentinel-2 and terrain data

A statistical comparison of six supervised classifiers
(using Kaggle dataset)



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October 2026

Content



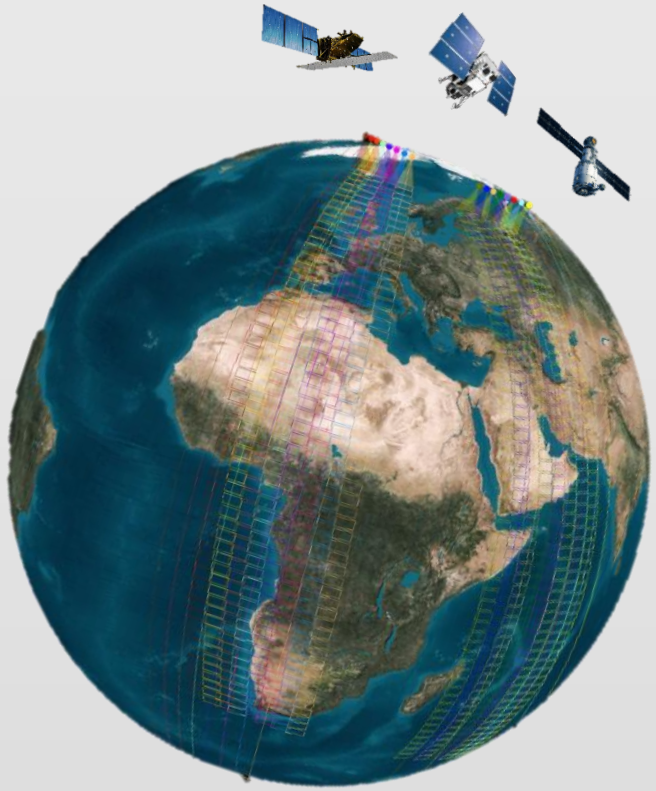
1 Problem and motivation

2 Data

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5 Limitations and Future Work



1 Problem and motivation



Fast landslide maps save lives — but landslides are small and rare

- Emergency response needs fast landslide maps
- **Challenge**: landslides are small and look like other bright, bare surfaces

1 How well can pixel-based classifiers detect landslides from multispectral and terrain data?

2 Which classifier is best — and are the differences real or just noise?

3 Where and why do all classifiers fail, and what would fix it?

2 Data



Landslide4Sense
(Kaggle)

3,799 labelled patches, 14 input layers

01

- **Patches:** 128 × 128 pixels at about 10 m resolution

02

- **Sentinel-2 (ESA):** 12 bands B1–B12, from visible light and red-edge to near- and short-wave infrared

03

- **ALOS PALSAR:** slope and elevation (DEM)

04

- **Four study areas:** Iburi (Japan), Kodagu (India), Gorkha (Nepal), Taiwan

05

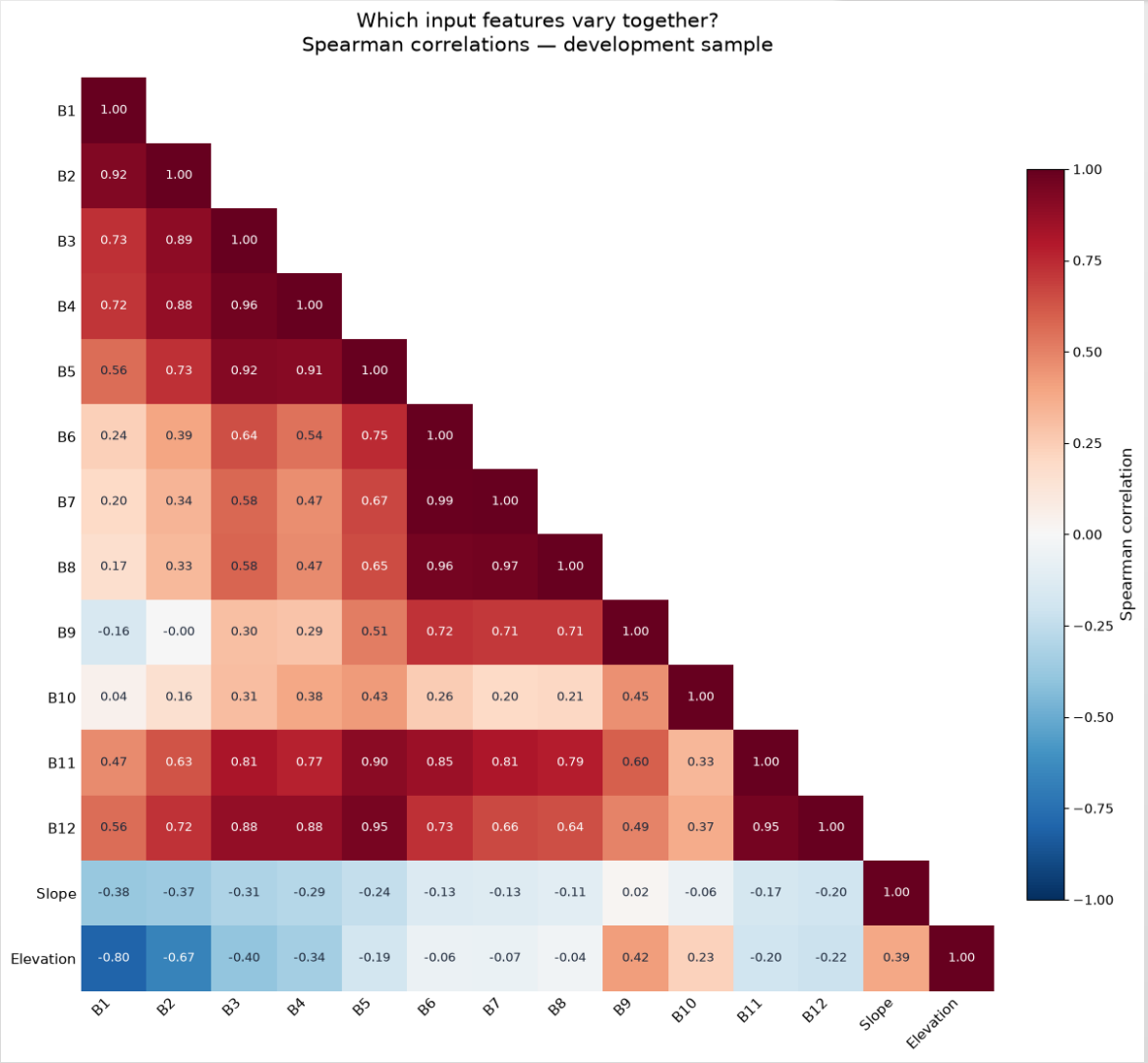
- **Split by image:** 3,039 development and 760 test patches, stratified by landslide presence

2.32%:
"of all pixels are landslide
(≈ 1 : 42 imbalance)"

2.0%:
"median landslide area per image that contains landslides"

2 Data

Data characteristics

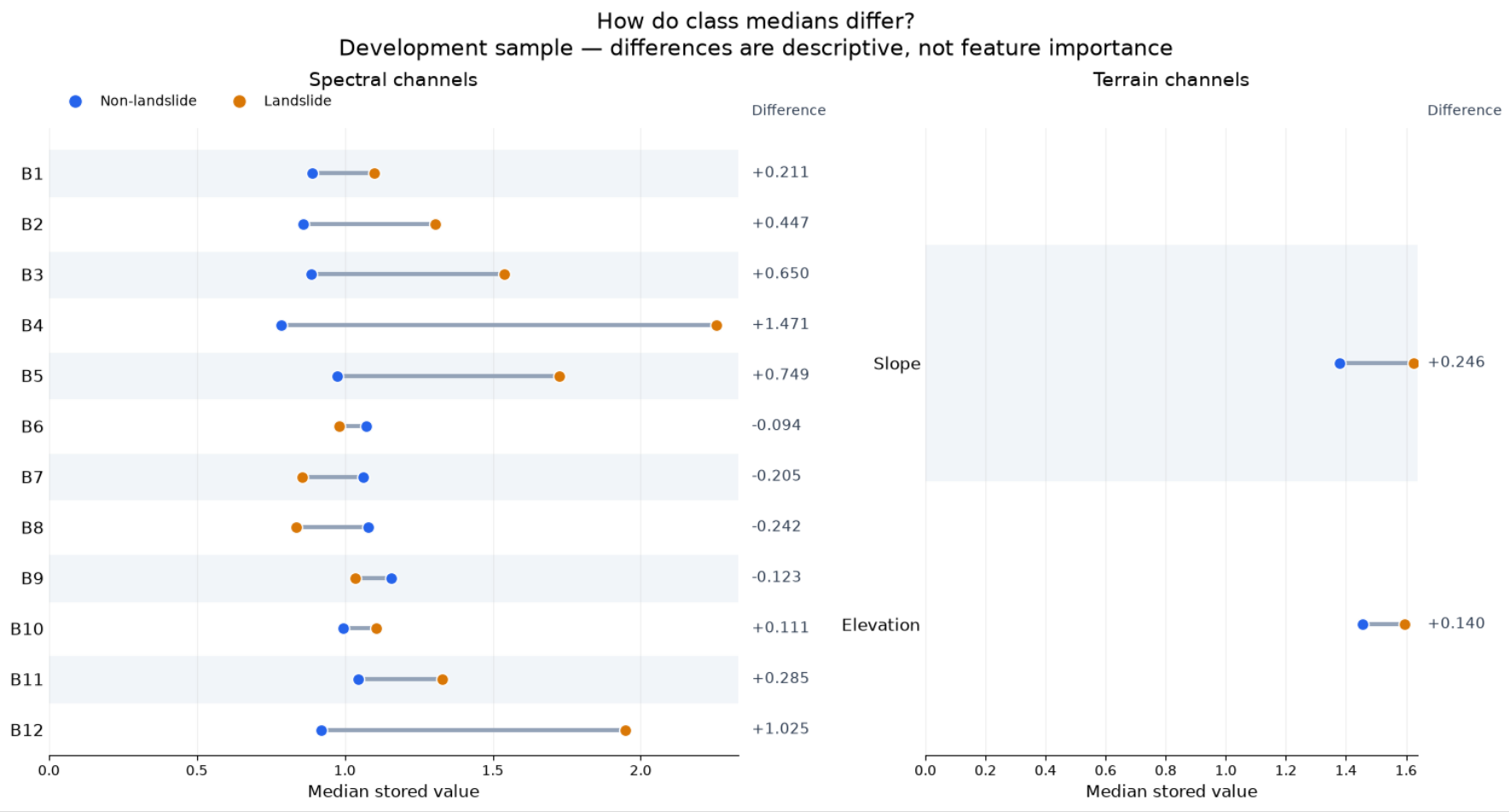


Many channels carry the same information

- **Strongly correlated band groups:** visible bands B2–B5 (ρ = 0.73–0.96), near-infrared B6–B8 (ρ ≥ 0.96), and SWIR B11–B12 (ρ = 0.95)
- **Terrain adds new information:** slope and elevation are only weakly related to most bands. The exception is elevation with B1 (ρ = −0.80).
- **Consequence:** correlated bands can replace each other in a model, so feature importance must be read for band groups, not single bands.

2 Data

Data characteristics



Landslides are bright in visible and SWIR bands

- Largest median differences: red B4 (+1.47), SWIR B12 (+1.03), red-edge B5 (+0.75).
- Near-infrared B6–B8 is slightly lower (vegetation loss). Landslides lie on steeper slopes.

3 Methodology and Analysis



Six classifiers

simple, linear → *more flexible, non-linear*

Linear models

one weighted sum of the inputs

Logistic regression — t = 0.10

14 standardised inputs · C = 0.1 · sigmoid calibration

LR + interactions — t = 0.11

+ 91 pairwise products (105 predictors) · C = 1

Interaction terms add +0.13 test F2: the problem is strongly non-linear

Single tree

a hierarchy of if–then rules

Decision tree — t = 0.92

depth 15 · 3,032 leaves · entropy · balanced class weights

Easy to read, but overfits: training F2 ≈ 0.83, validation only 0.70–0.74

Tree ensembles

many trees combined

Random forest — t = 0.17

bagging · 200 trees, depth 20 · 50% of features per split

HGB — t = 0.13

boosting · 150 trees · 63 leaves · learning rate 0.1

XGBoost — t = 0.14

boosting · 600 trees, depth 4 · learning rate 0.05 · L2 = 10

Decision threshold t (chosen by inner-CV F2): 0.10–0.17 for five of six models because landslides are rare — only the decision tree needs 0.92, as its balanced class weights push scores towards 1.

3 Methodology and Analysis



One fair protocol for all six classifiers

STEP 1

Split by image



3,039 development — 760 test

Stratified by landslide presence; the pixels of an image stay together.

STEP 2

Nested CV



5 outer × 3 inner folds, identical for every classifier.

STEP 3

Train and select

train — 512 / 1,024 px

evaluate — all 16,384 px

Settings, calibration and threshold chosen by inner-fold F2 only.

STEP 4

Lock and test

760 test images, predicted once

Fitted on all 3,039 images, then locked: no further tuning.

EVALUATION METRIC

$$F_2 = 5 \cdot TP \,/\, (5 \cdot TP + 4 \cdot FN + FP)$$

Recall counts 4× as much as precision: a missed landslide costs more than a false alarm.

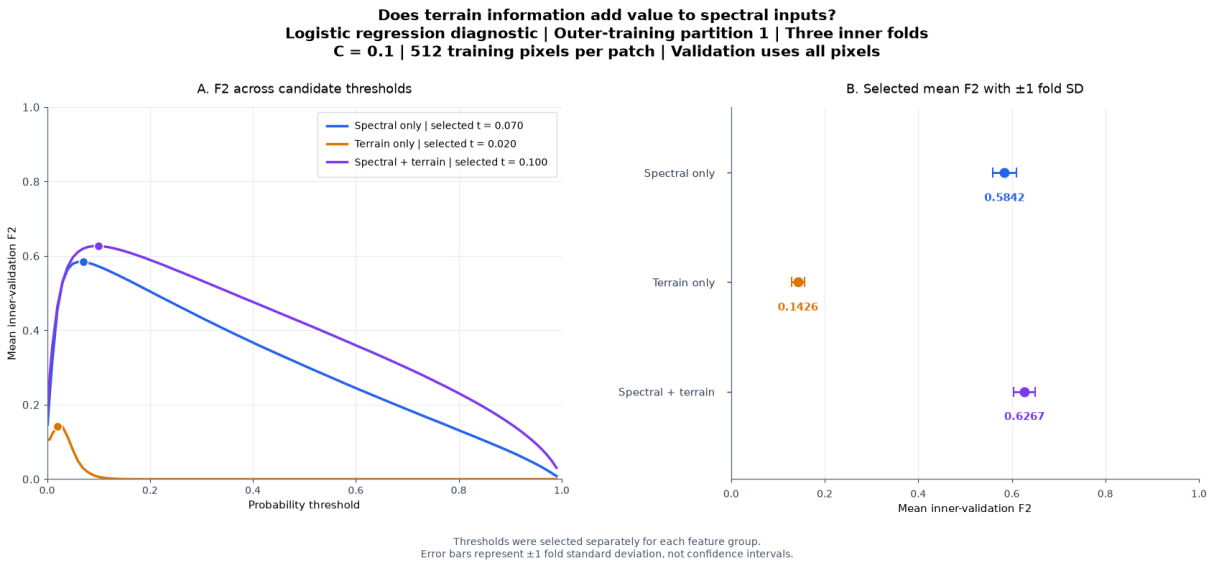
STATISTICAL COMPARISON · IMAGE = UNIT OF ANALYSIS

- Paired bootstrap
- Wilcoxon signed-rank
- Bonferroni
- Bayesian ROPE ±0.01 F2
- Cochran's Q · McNemar

3 Methodology and Analysis

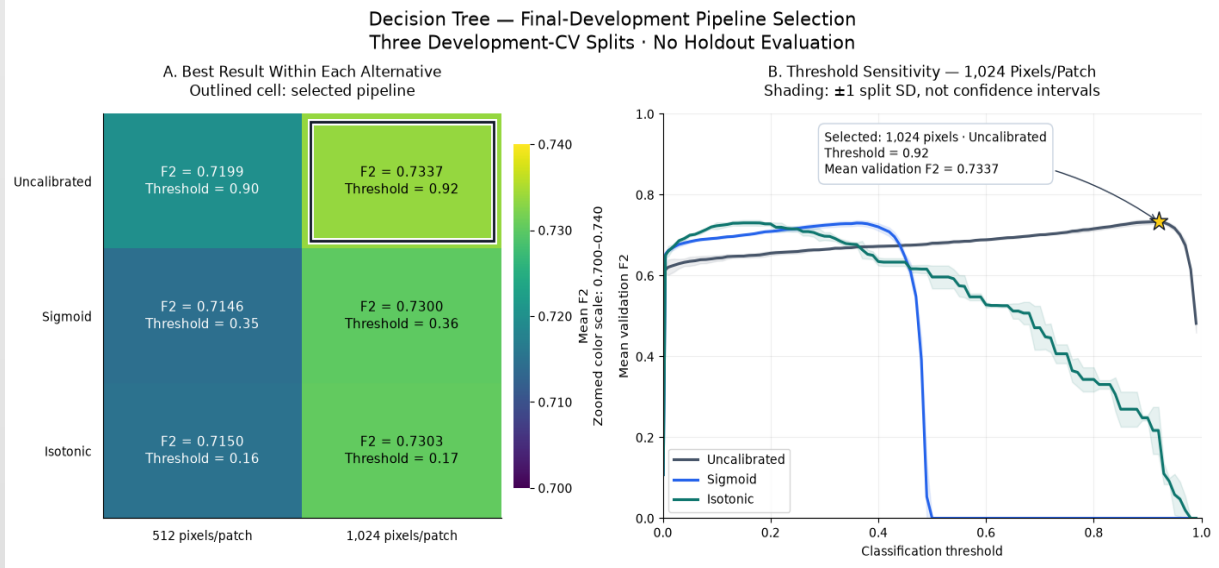
Model selection: features, calibration and threshold

1 · DO TERRAIN LAYERS HELP? (LOGISTIC REGRESSION)



Spectral 0.584 + terrain → 0.627 inner F2: all 14 channels kept for every model.

2 · CALIBRATION × THRESHOLD (DECISION TREE)



Uncalibrated tree at threshold 0.92 wins (F2 0.734); the threshold matters more than calibration.

Setting	LR	LR + interactions	Decision tree	Random forest	HGB	XGBoost
Search space	C, class weight, pixels	C (0.1 / 1 / 10)	25 random configs	8 configs + checks	6 candidates (114 fits)	3 candidates
Calibration	**sigmoid (Platt)**	none	none	none	none	none
Threshold	0.10	0.11	**0.92**	0.17	0.13	0.14
Final fit	*not timed*	*not timed*	27 s	**8–11 min**	63 s	74 s

4 Results



Ensembles lead with a test F2 of about 0.77



SECONDARY TEST · PIXEL LEVEL

Classifier	Precision	Recall	IoU	AP
Random forest	**.535**	.860	**.492**	**.780**
HGB	.527	**.864**	.487	.774
XGBoost	.530	.860	.488	.773
LR + interactions	.511	.838	.465	.738
Decision tree	.529	.796	.466	.697
Logistic regression	.360	.741	.320	.504

≈ 86%
of landslide pixels found by the ensembles

1 in 2
flagged pixels is a false alarm

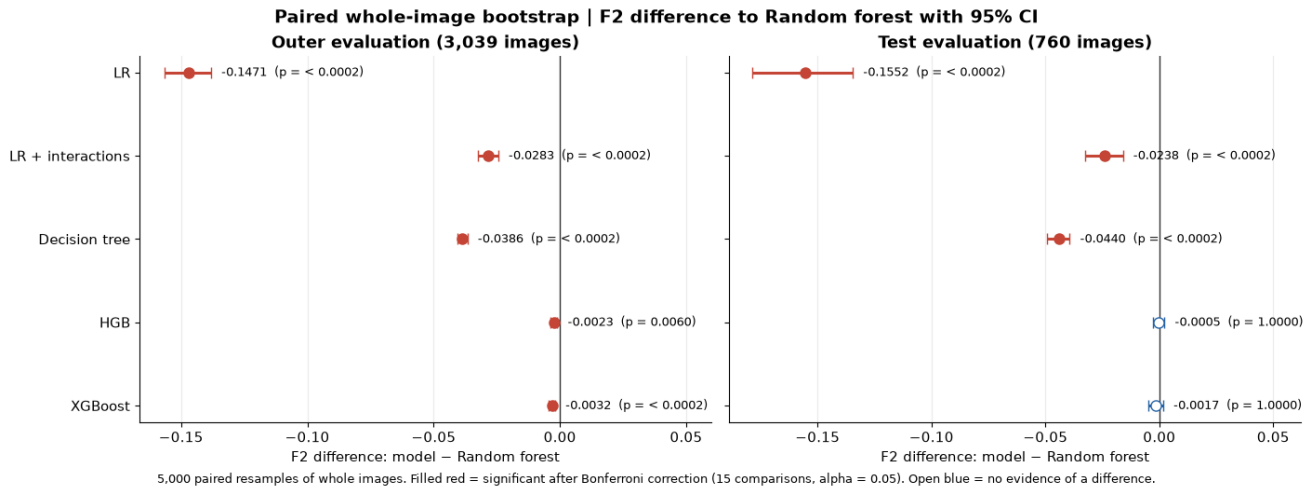
+0.13
F2 from interaction terms in LR

No overfitting: test F2 is only 0.005–0.018 below outer F2 for every model (not significant).

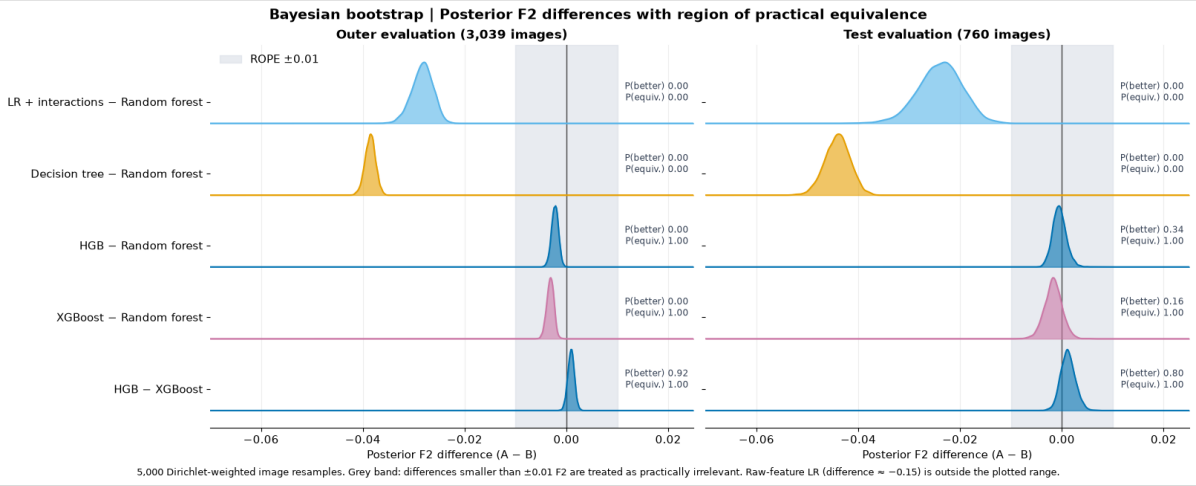
4 Results

The three ensembles are practically equivalent

PAIRED WHOLE-IMAGE BOOTSTRAP · ΔF2 TO RANDOM FOREST



BAYESIAN POSTERIOR · REGION OF PRACTICAL EQUIVALENCE ±0.01



Four clear groups

Ensembles > LR + interactions > decision tree > LR — all significant after Bonferroni (15 pairs).

Tiny gaps at the top

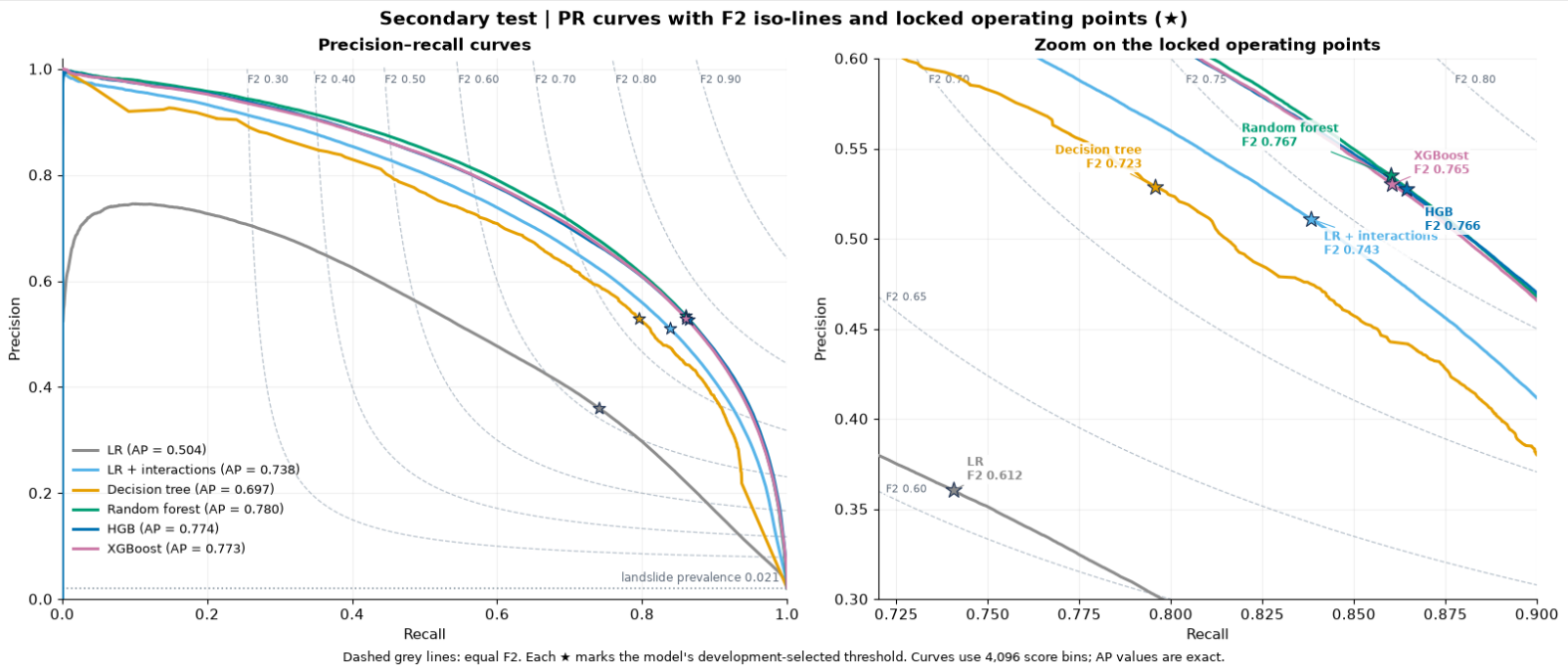
Outer: RF ahead of HGB / XGBoost by only +0.002 / +0.003 F2. Test: −0.0005 and −0.0017, both p = 1.0.

Bayesian verdict

P(|ΔF2| < 0.01) = 1.00 for every ensemble pair, in both evaluations: practically equivalent.

4 Results

The ensembles rank pixels almost identically



AVERAGE PRECISION · TEST

Classifier	AP
Random forest	**.780**
HGB	.774
XGBoost	.773
LR + interactions	.738
Decision tree	.697
Logistic regression	.504

Overlapping curves

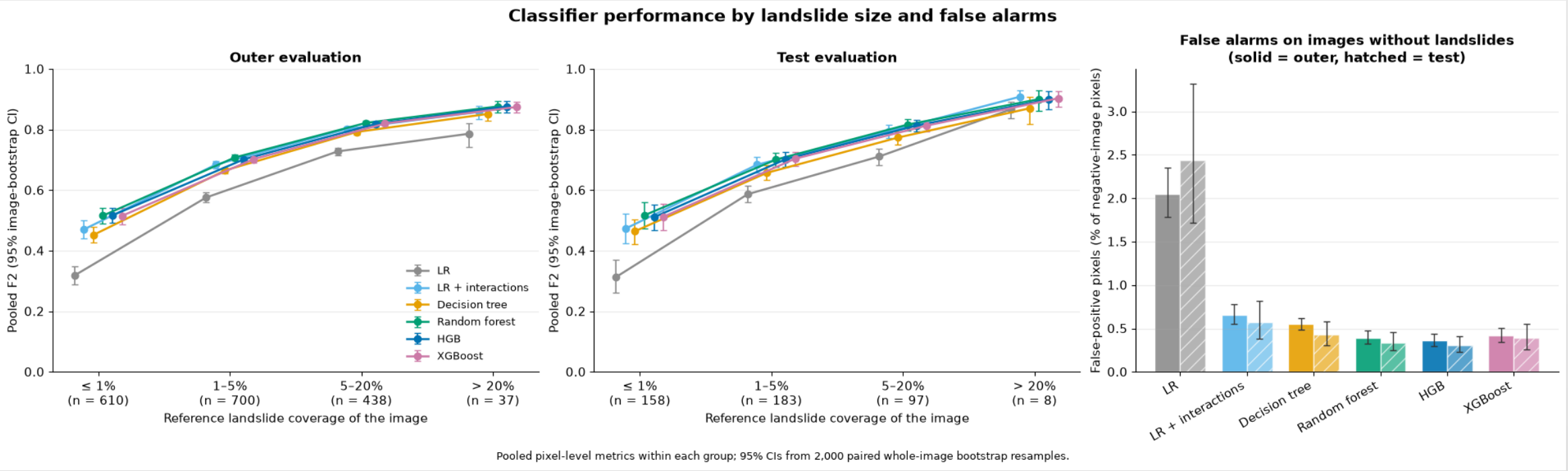
Random forest, HGB and XGBoost lie on top of each other over the full recall range — not only at the chosen threshold.

Same operating region

The ensembles and LR + interactions operate at recall ≈ 0.84–0.86 and precision ≈ 0.51–0.54 (★), close to the same F2 iso-line.

4 Results

Landslide size matters more than the classifier



0.52 → 0.88

F2 of the best model from ≤ 1 % to > 20 % landslide coverage per image

≈ 2×

larger lead of the ensembles for small landslides (difference-in-differences)

2 % vs < 0.5 %

of pixels falsely flagged on landslide-free images: raw LR vs. ensembles

4 Results

Image by image: high detection, few clean images

PER-IMAGE F2 ON IMAGES WITH LANDSLIDES

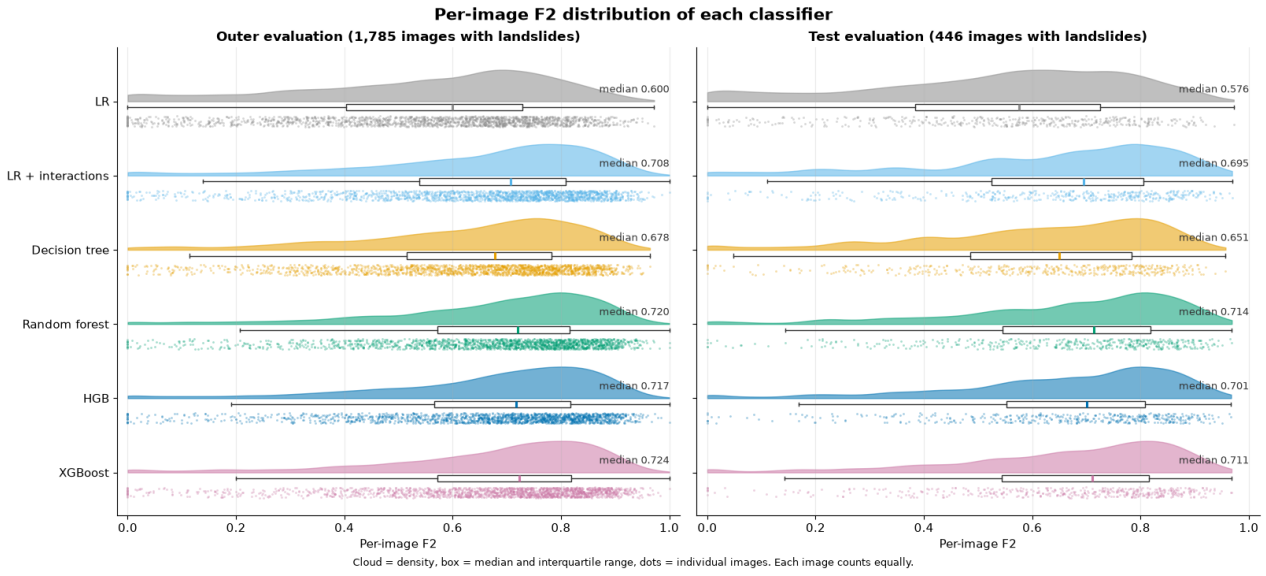


IMAGE-LEVEL OUTCOMES · TEST SET

Classifier	Landslide images detected	Landslide-free images clean	Median F2
Random forest	**91.5 %**	32.5 %	**0.714**
HGB	90.6 %	33.4 %	0.701
XGBoost	91.0 %	**35.4 %**	0.711
LR + interactions	86.5 %	25.8 %	0.695
Decision tree	85.4 %	19.4 %	0.651
Logistic regression	78.5 %	17.2 %	0.576

Detected = at least 50 % of the landslide pixels found. Clean = not a single false-positive pixel.

Most landslides are found

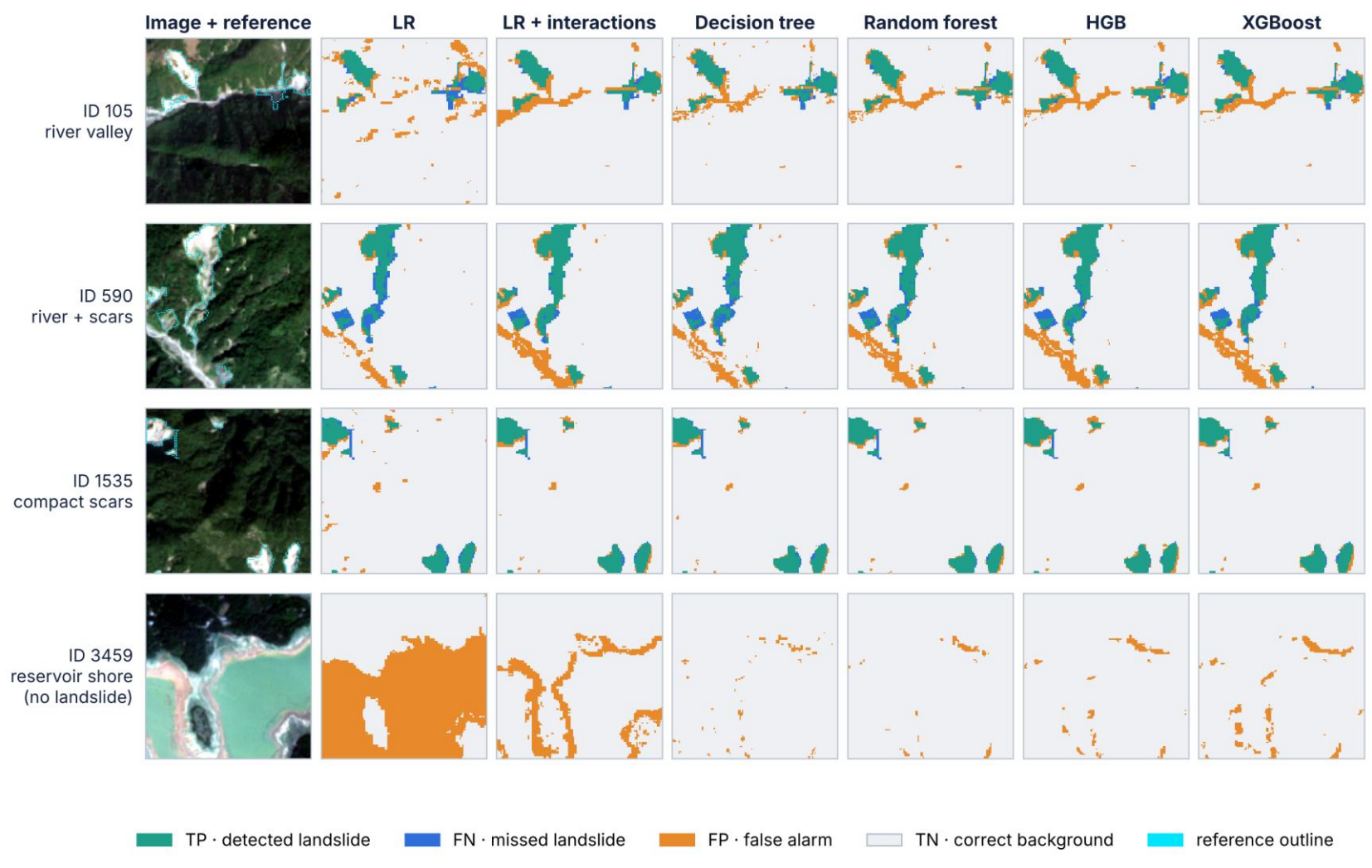
Ensembles detect at least half of the landslide in about 91 % of images, but every model has a long tail of poorly detected images — mostly tiny landslides.

A small trade-off between ensembles

Random forest detects the most landslide images; XGBoost leaves the most landslide-free images completely clean.

4 Results

Every classifier makes the same key mistakes



Shared false alarms

River channels in ID 105 and 590 are flagged (orange) by all six classifiers.

Shared misses

The square scar in ID 590 is missed (blue) by every model.

Model-specific errors

The reservoir shore (ID 3459) fools mainly LR; the ensembles stay almost clean.

Whole test set

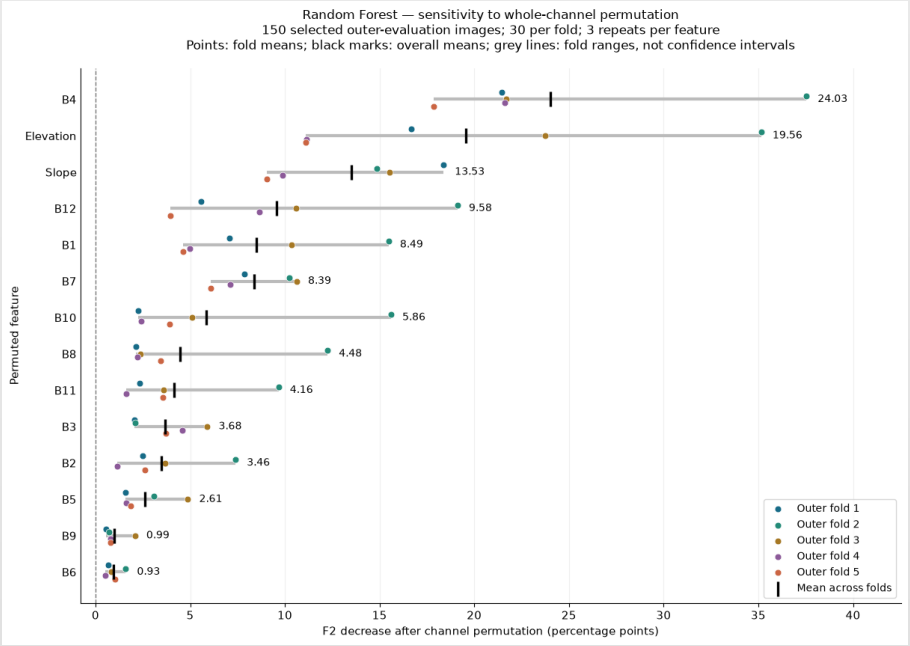
8.0 % of landslide pixels are missed by all six models — about a quarter of all pixels missed by any model.

4 Results

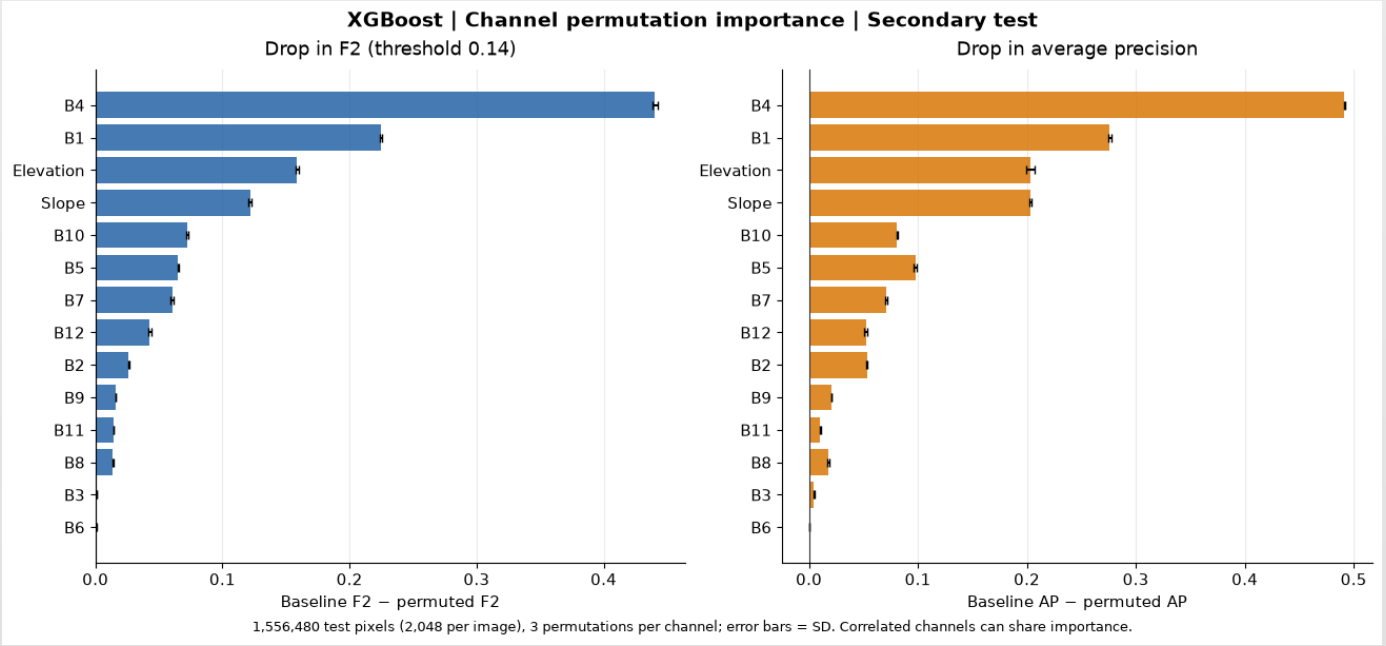


All models rely mainly on the red band B4

RANDOM FOREST · PERMUTATION IMPORTANCE



XGBOOST · PERMUTATION IMPORTANCE (TEST SET)

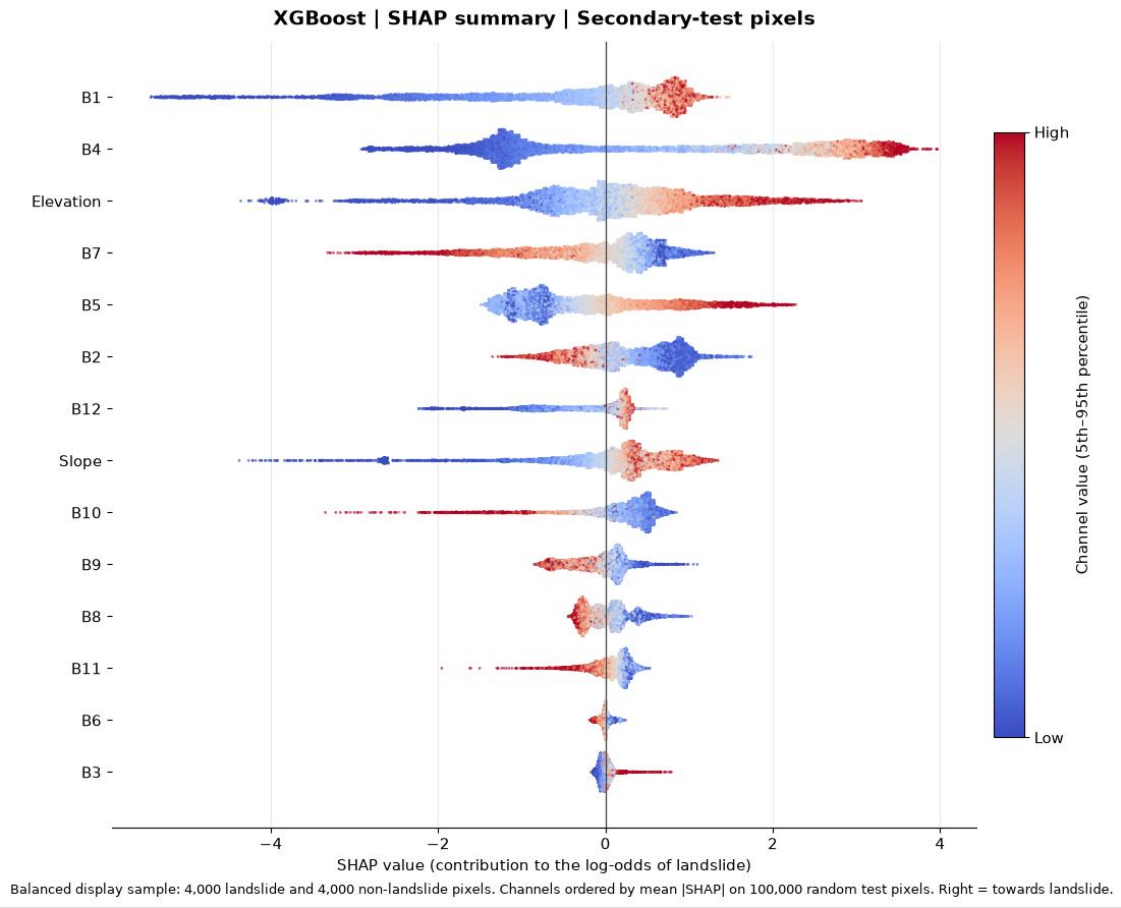


Top channels (permutation)	1st	2nd	3rd	4th
Decision tree	**B4**	Elevation	B7	Slope
Random forest	**B4**	Elevation	Slope	B12
HGB	**B4**	B1	Elevation	Slope / B7
XGBoost	**B4**	B1	Elevation	Slope

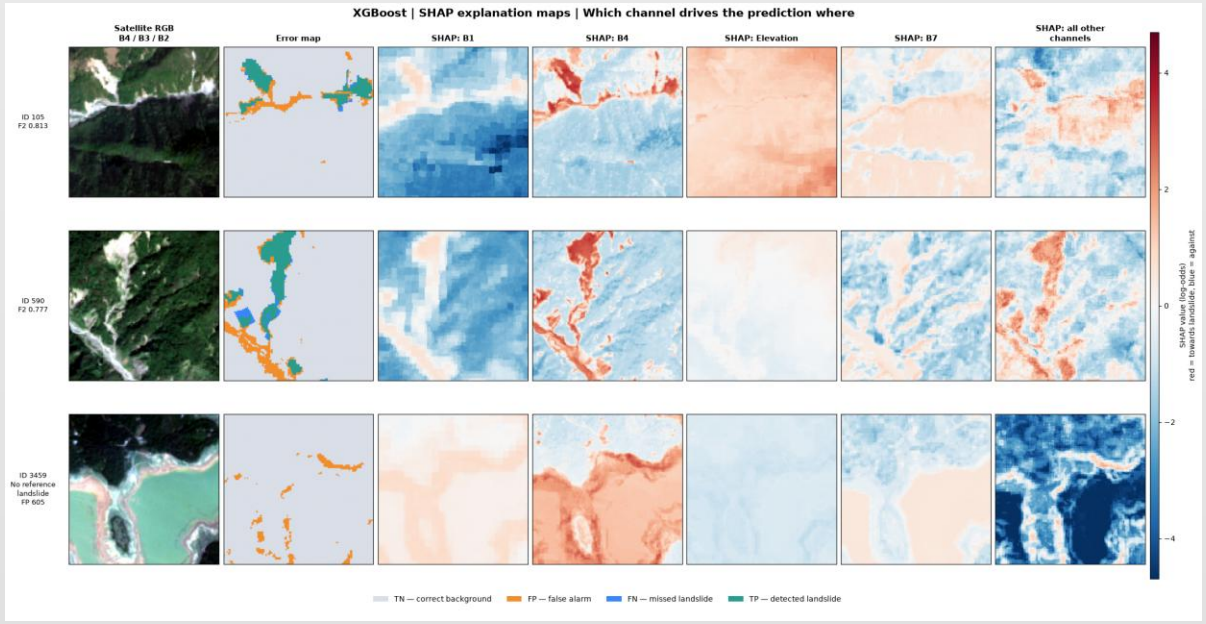
4 Results

Why errors happen: riverbeds look like landslides in B4

SHAP SUMMARY · XGBOOST



SHAP EXPLANATION MAPS · THREE TEST IMAGES



B4 = main evidence

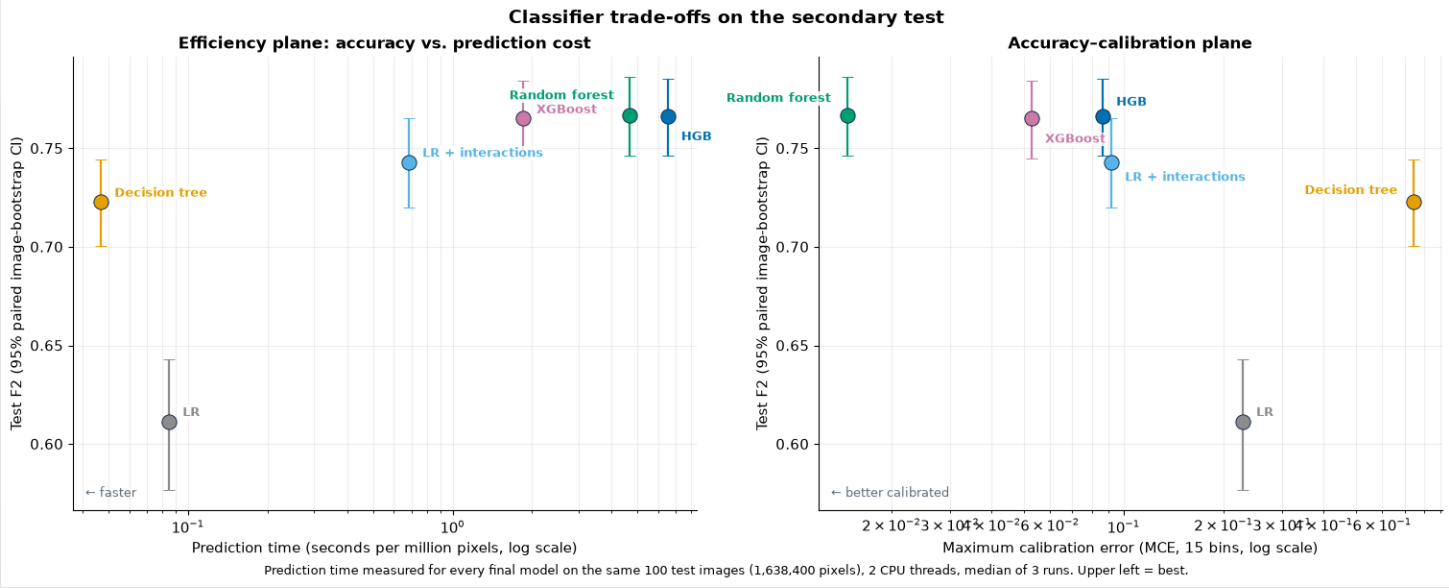
High red reflectance adds about +2.1 log-odds on landslide pixels; low B1 rules landslides out.

Physical failure mode

Riverbeds and bright shores push B4 the same way as real scars → shared false alarms.

4 Results

Equal accuracy, different reliability and speed



Classifier	ECE	MCE	s / M px	Final fit
Random forest	**0.0010**	**0.015**	4.69	8–11 min
HGB	0.0018	0.087	6.53	63 s
XGBoost	0.0011	0.053	**1.85**	74 s
LR + interactions	0.0027	0.092	0.69	*not timed*
Decision tree	**0.0426**	**0.746**	0.05	27 s
Logistic regression	0.0042	0.228	0.09	*not timed*

ECE / MCE: expected / maximum calibration error (15 bins). Prediction time in seconds per million pixels.

Random forest

Best-calibrated probabilities (MCE 0.015) — the choice for probability and risk maps.

XGBoost

2.5× faster than RF, 3.5× faster than HGB; nested search in ≈ 73 min instead of ≈ 40 h.

Decision tree

Scores are not probabilities (MCE 0.75): balanced class weights push them towards 1.

5 Limitations and Future Work



LIMITATION	NEXT STEP
Single pixels only No shape or texture: riverbeds look like landslides.	**Add spatial context** U-Net / transformer segmentation, or texture features.
Small landslides F2 ≈ 0.52 when landslides cover ≤ 1 % of an image.	**Change detection** Pre-/post-event change and multi-scale inputs reveal small scars.
Optical data only Clouds can hide the area right after an event.	**Add Sentinel-1 radar** SAR sees through clouds; fuse it with Sentinel-2.
Four regions only Same events in development and test: no proof of transfer.	**Region-wise validation** Train on some regions, test on completely unseen ones.
Test set already inspected Viewed during the LR analysis, so it is a secondary test.	**Untouched test region** Lock a new region away and evaluate it only once.

APPLICATIONS Rapid-response mapping · Landslide inventories · Susceptibility & risk maps · Early-warning support

Thanks for Your Attention

